

Exact CRT Tensoring for an Odd-Level Metaplectic Hénon Family

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Abstract

The odd-level quantization of the toral Hénon matrix $A = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}$ is defined separately at every odd modulus, but a per-level theorem does not fix its cross-level phases. We enlarge the family by a unit additive-character parameter and prove exact Chinese-remainder tensor identities for the Weyl, Fourier, chirp, and evolution operators at every pair of coprime odd levels. The induced local characters contain inverse cofactor scalings. Iterating the identity gives a finite-factor tensor law independent of binary bracketing. Exact certificates cover 1,067,332 modular cases. At levels 3 and 5, the naive tensor of the standard local characters has exponent 8 rather than 1 modulo 15, so an overall scalar cannot repair it. The result is a cross-level quantization theorem, not a target spectral or arithmetic claim.

1 The cross-level question

For an odd $r \geq 3$, write $\omega_r = e^{2\pi i/r}$ and let $h_r = 2^{-1} \pmod{r}$. The earlier per-level family used

$$Q_r|x\rangle = \omega_r^x|x\rangle, \quad P_r|x\rangle = |x+1\rangle,$$

$$W_r(q, p) = \omega_r^{-h_r q p} Q_r^q P_r^p, \quad (\mathcal{F}_r)_{xy} = r^{-1/2} \omega_r^{xy}, \quad (C_r)_{xx} = \omega_r^{3h_r x^2}, \quad U_r = C_r \mathcal{F}_r^{-1}.$$

It gives exact Egorov covariance at each odd level. The unresolved question is whether the canonical residue-basis identification at a product level respects these phases. Dimension counting alone cannot answer it.

2 Generalized additive characters

Let $c \in (\mathbb{Z}/r\mathbb{Z})^\times$. Define

$$W_{r,c}(q, p) = \omega_r^{-ch_r q p} Q_r^{cq} P_r^p,$$

$$(\mathcal{F}_{r,c})_{xy} = r^{-1/2} \omega_r^{cxy}, \quad (C_{r,c})_{xx} = \omega_r^{3ch_r x^2}, \quad U_{r,c} = C_{r,c} \mathcal{F}_{r,c}^{-1}.$$

The original family is the fiber $c = 1$.

Proposition 1 (unchanged classical action). *For every odd $r \geq 3$ and unit c , $U_{r,c}$ is unitary and*

$$U_{r,c} W_{r,c}(q, p) U_{r,c}^{-1} = W_{r,c}(3q - p, q).$$

Thus one quantum step still implements one application of A .

Proof. The root ω_r^c is primitive, so finite Fourier orthogonality proves that $\mathcal{F}_{r,c}$ is unitary. The chirp is diagonal unitary. Direct generator conjugation gives $\mathcal{F}_{r,c}^{-1} Q_r^c \mathcal{F}_{r,c} = P_r$ and $\mathcal{F}_{r,c}^{-1} P_r \mathcal{F}_{r,c} = Q_r^{-c}$. After chirp conjugation these become $W_{r,c}(3, 1)$ and $W_{r,c}(-1, 0)$. The symmetric half-phases cancel in a general Weyl product, giving the displayed identity without a scalar. $\square$

3 Exact two-factor CRT identity

Let $M, N > 1$ be coprime odd integers and $L = MN$. Put

$$a = N^{-1} \pmod{M}, \quad b = M^{-1} \pmod{N},$$

and, for a unit $c \pmod{L}$,

$$c_M = (c \bmod M)a \pmod{M}, \quad c_N = (c \bmod N)b \pmod{N}.$$

The canonical unitary is

$$J_{M,N}|x \bmod L\rangle = |x \bmod M\rangle \otimes |x \bmod N\rangle.$$

Theorem 1 (exact CRT tensoring). *Under this identification,*

$$\begin{aligned} J\mathcal{F}_{L,c}J^{-1} &= \mathcal{F}_{M,c_M} \otimes \mathcal{F}_{N,c_N}, & JC_{L,c}J^{-1} &= C_{M,c_M} \otimes C_{N,c_N}, \\ JW_{L,c}(q,p)J^{-1} &= W_{M,c_M}(q_M,p_M) \otimes W_{N,c_N}(q_N,p_N), & JU_{L,c}J^{-1} &= U_{M,c_M} \otimes U_{N,c_N}. \end{aligned}$$

All four identities are exact.

Proof. The CRT idempotents $e_M = Na$ and $e_N = Mb$ reconstruct $x = e_M x_M + e_N x_N \pmod{L}$. Their cross product vanishes modulo L , and

$$\frac{xy}{L} \equiv \frac{ax_M y_M}{M} + \frac{bx_N y_N}{N} \pmod{1}.$$

Multiplication by c gives the two induced characters. Since h_L reduces to h_M and h_N , the same calculation factors the chirp and the symmetric Weyl phase. Shifting x by p shifts both residue coordinates. These facts give the first three identities entry by entry; multiplying the chirp and inverse Fourier identities gives the fourth. Finally, $L^{-1/2} = M^{-1/2}N^{-1/2}$ with positive roots, so no normalization phase is left over. $\square$

4 Finite-factor coherence

Let $r_1, \dots, r_k > 1$ be pairwise coprime and odd, and put $L = \prod_j r_j$. For a unit $c \pmod{L}$, define

$$c_j = (c \bmod r_j)(L/r_j)^{-1} \pmod{r_j}.$$

Theorem 2 (bracket independence). *Under the canonical residue-basis identification,*

$$JU_{L,c}J^{-1} = \bigotimes_{j=1}^k U_{r_j, c_j}.$$

The local characters and the operator identity are independent of the binary parenthesization used to construct the CRT map.

Proof. Apply Theorem 1 recursively. If $L = RS$ and $r_j \mid R$, the first split contributes S^{-1} and the split inside R contributes $(R/r_j)^{-1}$. Their product is $(L/r_j)^{-1} \pmod{r_j}$. Induction gives the direct coefficient above for every bracketing, and the operator identities compose exactly. $\square$

5 Certificate, control, and boundary

The deterministic receipt exhausts eight two-factor systems and four three-factor systems. It checks 13,520 Fourier kernels, 306 chirp entries, 13,520 unitary kernels, 658,314 Weyl basis actions, and 381,672 three-factor unitary kernels. A four-factor system compares left, right, and balanced bracketings. An independent implementation reconstructs every case-ledger hash; a symbolic path checks the same congruences through a different code path.

The inverse coefficients cannot be erased. At $(M, N) = (3, 5)$ one has $a = b = 2$. For Fourier entry $x = y = 1$, the level-15 exponent is 1; the naive standard tensor gives $5 + 3 = 8$, whereas the induced tensor gives $5 \cdot 2 + 3 \cdot 2 = 16 \equiv 1 \pmod{15}$. The output row $x = 0$ agrees exactly and therefore fixes any proposed projective scalar to one, so the mismatch is not a scalar anomaly.

The theorem excludes noncoprime factors because the residue map is then not a product-ring bijection. It also excludes even factors under this literal $qp/2$ convention; other even-level Weil conventions are not ruled out. No correction back to standard local characters, semiclassical trace match, target divisor, Euler factor, root number, automorphy, or Hilbert–Pólya operator is asserted.

The strict Route-A tuple is

$$(A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION),$$

and Route B is unauthorized.

6 Conclusion

The inverse-scaled additive characters turn the odd-level Hénon quantization into an exact CRT tensor family, with finite-factor coherence and the original one-step clock. The standard-character shortcut fails. A separate problem may ask whether natural local intertwiners can remove the induced twists; no such structure is inferred here.